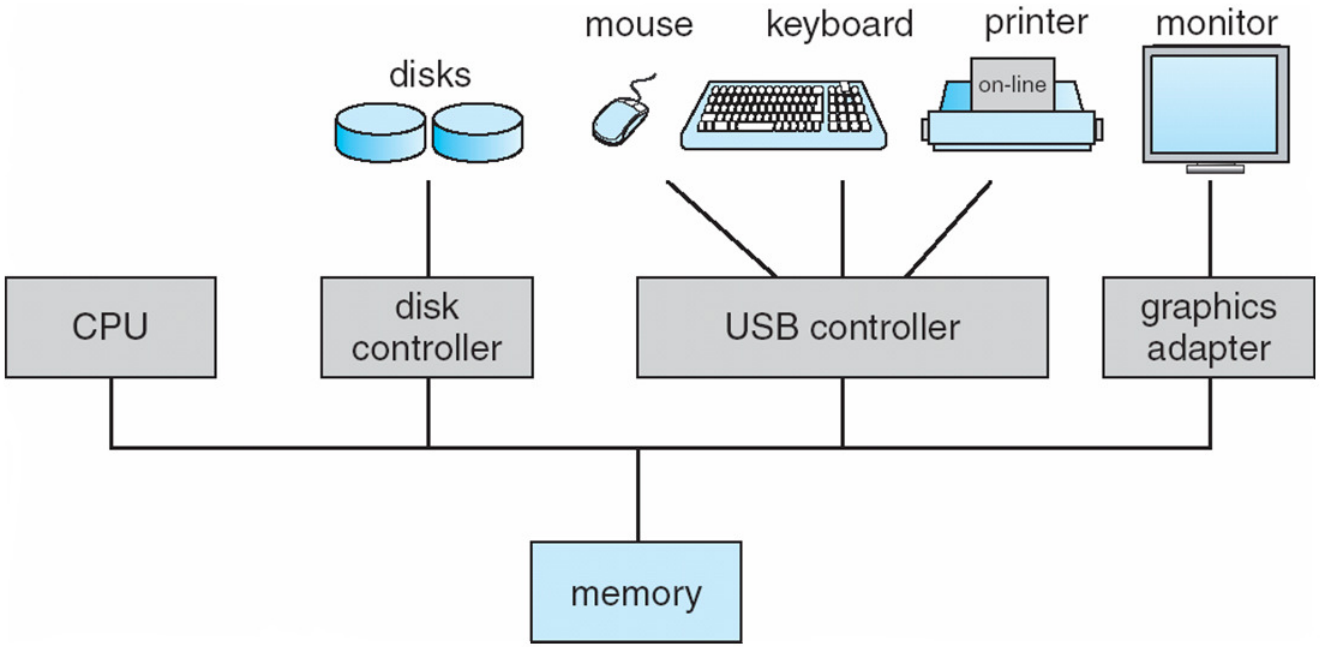
I/O Systems

Introduction

- I/O management is a major component of operating system design and operation
  - Important aspect of computer operation
  - I/O devices vary greatly:
    - Machine Readable, Human Readable, Communication
  - Various methods to control them
  - Performance management
  - New types of devices frequent
  - Device Controllers



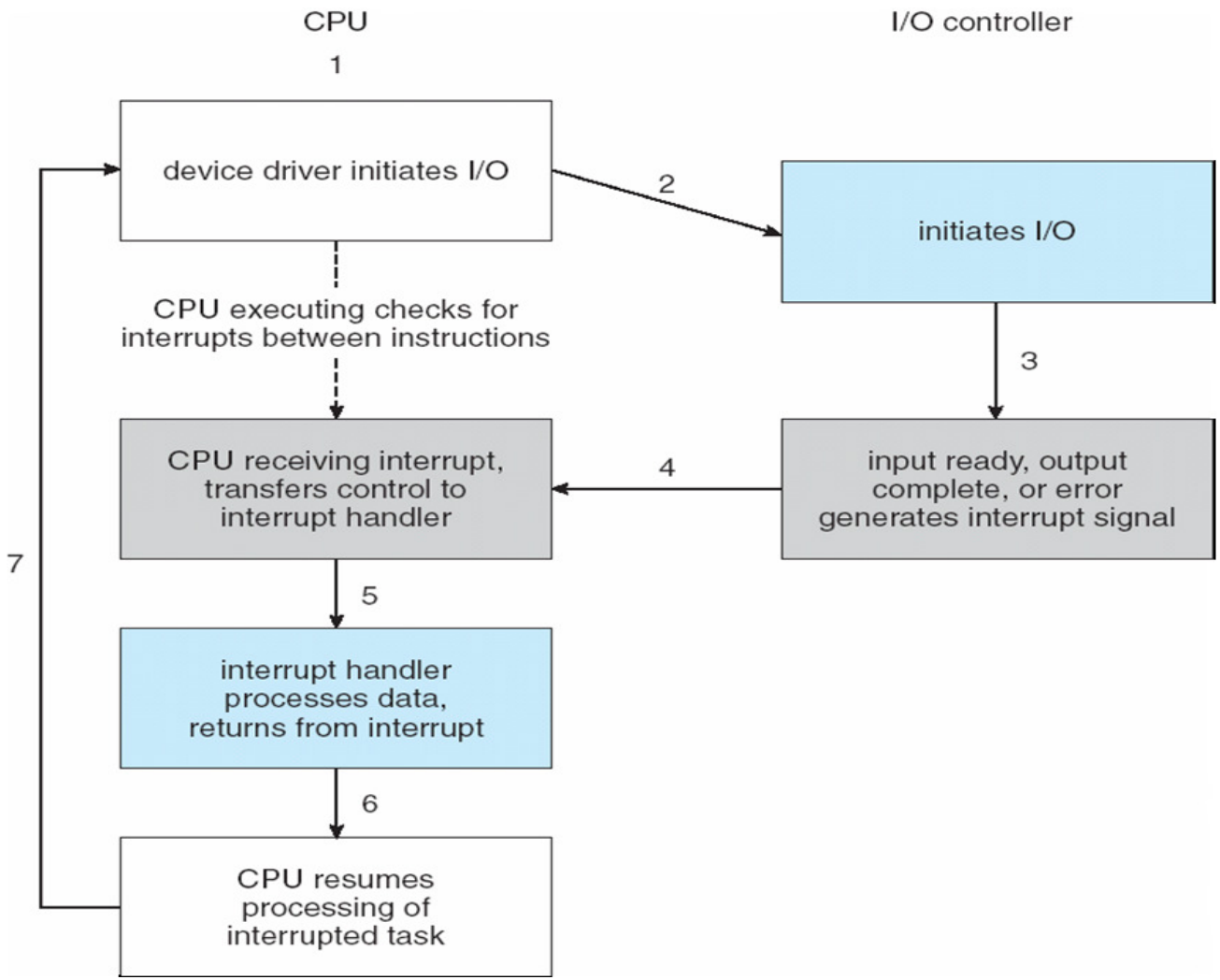
Introduction

- I/O instructions control devices – privileged instructions
- Devices controllers usually have registers where device driver places commands, addresses, and data to write, or read data from registers after command execution
  - Data-in register, data-out register, status register, control register
- Devices have addresses, used by
  - Direct I/O instructions/Isolated addresses
  - Memory-mapped I/O

I/O Operations

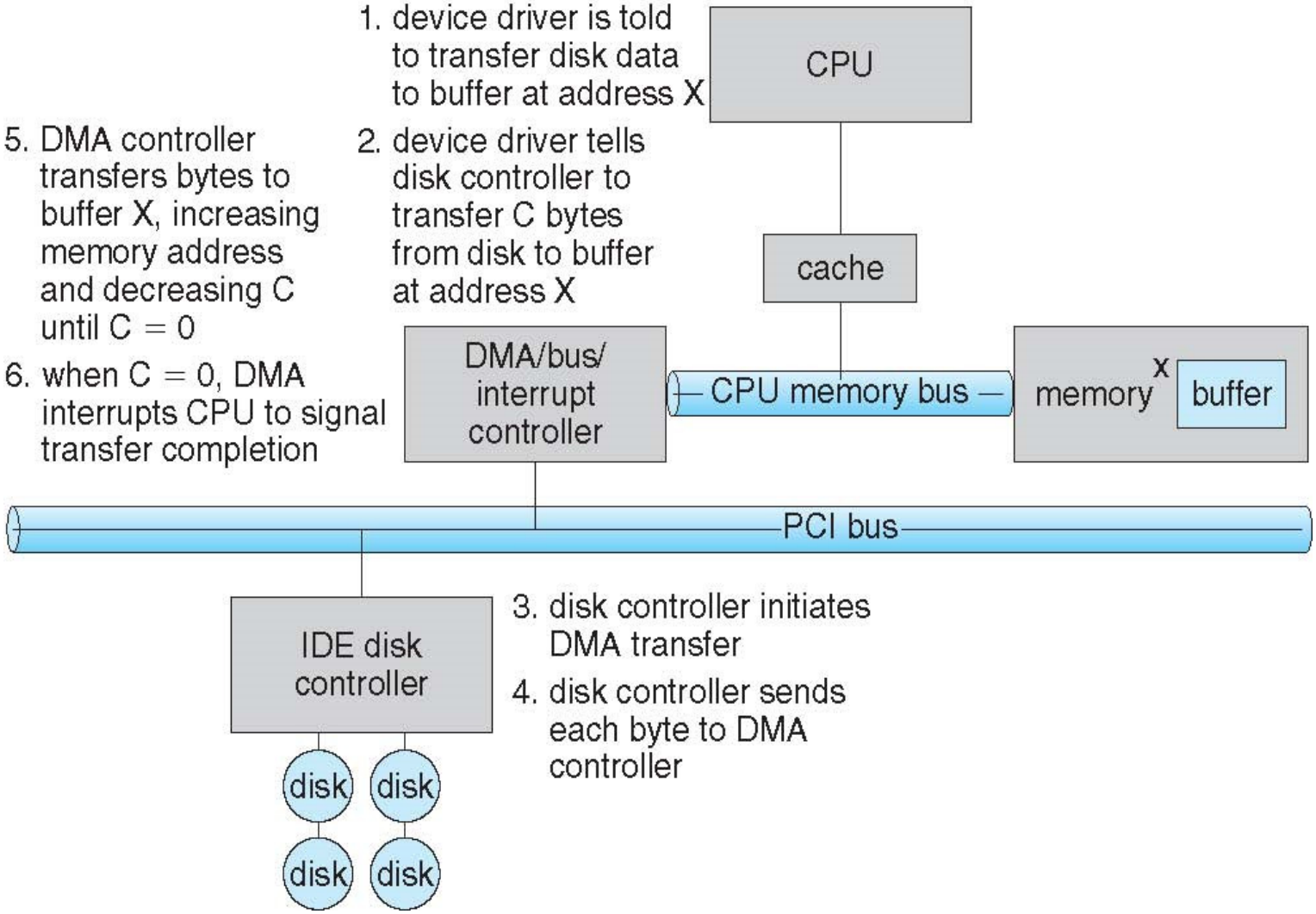
- Programmed I/O
  - CPU polls
  - Busy waiting
- Interrupt Driven I/O
  - Daisy Chaining
    - Interrupt Vector
  - Interrupt Controller Hardware
    - Maskable / Unmaskable interrupts
- DMA (Direct Memory Access)
  - DMA controller
  - Burst Transfer / Cycle stealing

Interrupt-Driven I/O Cycle

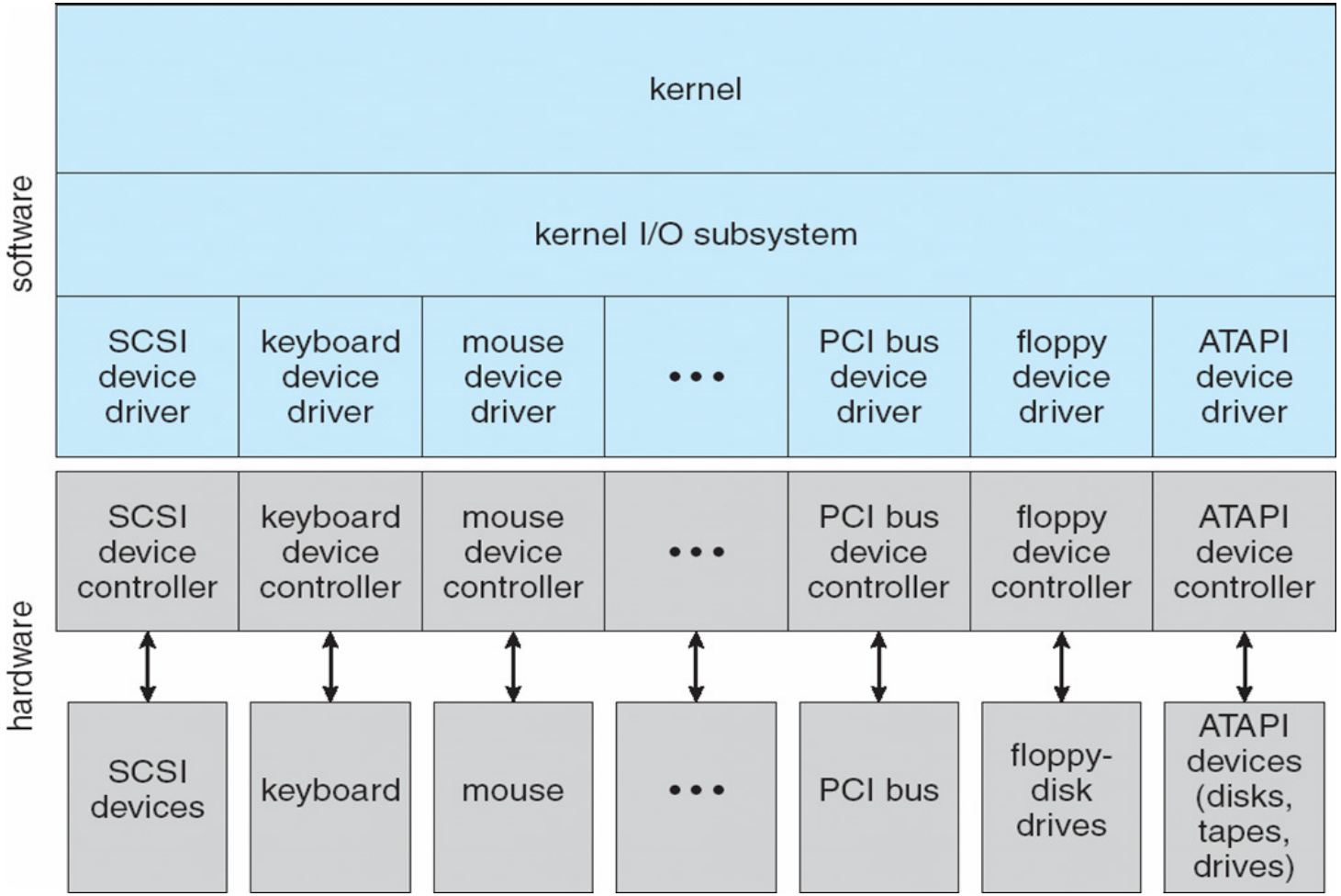


Direct Memory Access

- Bypasses CPU to transfer data directly between I/O device and memory - Requires **DMA** controller
- OS writes DMA command :
  - Source and destination addresses, Read or write mode, Count of bytes, Writes location of command block to DMA controller
  - Bus mastering of DMA controller – grabs bus from CPU
  - When done, interrupts to signal completion



Kernel I/O Structure



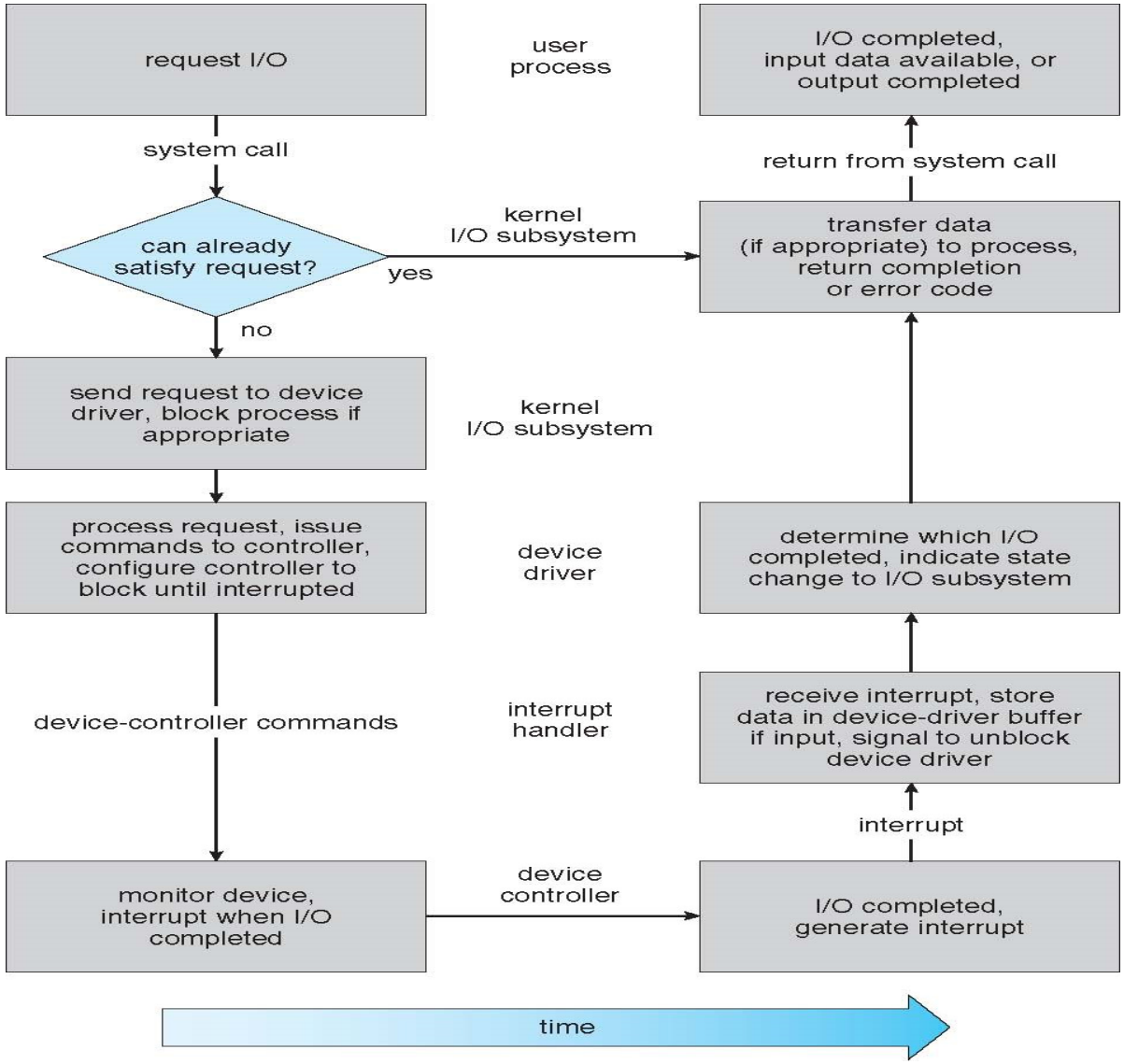
Application I/O Interface

- I/O system calls encapsulate device behaviors in generic classes
- Device-driver layer hides differences among I/O controllers from kernel
  - New devices talking already-implemented protocols need no extra work
  - Each OS has its own I/O subsystem structures and device driver frameworks
  - Making the I/O subsystem independent of the hardware simplifies the job of OS developers.
- Devices vary in many dimensions
  - Character-stream or block
  - Sequential or random-access
  - Synchronous or asynchronous (or both)
  - Sharable or dedicated
  - Speed of operation
  - read-write, read only, or write only

Kernel I/O Subsystem

- Kernels provide many services to I/O.
  - Scheduling
  - Buffering
  - Caching
  - Spooling
  - Device reservation
  - I/O Protection
  - Error Handling

I/O Requests to Hardware Operations



Reference

- "Operating System Concepts", Silberschatz, Galvin, Gagne, 9th Edition, Wiley India.